Multiform Ensemble Self-Supervised Learning for Few-Shot Remote Sensing Scene Classification

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***Abstract*— Self-supervised learning is an effective way to solve model collapse for few-shot remote sensing scene classification (FSRSSC). However, most self-supervised contrastive learning auxiliary tasks perform poorly on the high interclass similarity problem in FSRSSC. Furthermore, it is time-consuming and computationally expensive to obtain the best combination among numerous self-supervised auxiliary tasks. In practical applications, we may encounter difficulties in remote sensing data acquisition and labeling, while most FSRSSC studies only focus on the former. To alleviate the above problems, we propose a multiform ensemble self-supervised learning (MES^{2}L) framework for FSRSSC in this article. Based on the transfer learning-based few-shot scheme, we design a novel global–local contrastive learning auxiliary task to solve the low interclass separability problem. The self-attention mechanism is designed in the local contrast features to investigate the intrinsic associations between different remote sensing scene objectives. We also present a multiform ensemble enhancement (MEE) training method. Ensemble enhancement involves the concatenation of features extracted from different backbones trained by a combination of multiform self-supervised auxiliary tasks. MEE can not only be regarded as a more straightforward alternative to knowledge distillation but also can achieve an effective compromise between expensive computational cost and classification accuracy. In addition, we provide two scene classification schemes of inductive and transductive settings, corresponding to solving the difficulties of remote sensing data acquisition and labeling. The proposed network achieves state-of-the-art results on three benchmark FSRSSC datasets. The potential of the MES^{2}L framework is also demonstrated in combination with classical metalearning-based and metric learning-based few-shot algorithms.**

***Index Terms*— Few-shot learning, global–local contrastive learning, remote sensing scene classification, self-supervised learning.**

I. Introduction

SCENE classification aims to assign semantic labels to remote sensing images of different categories [1], [2], [3]. With the rapid development of high-quality sensors and deep learning algorithms, the labeling of massive data has become a challenging problem [4], [5], [6]. Few-shot scene classification, which aims to identify unseen remote sensing images with only a small number of labeled samples, has been extensively studied based on metalearning and transfer learning [7], [8]. The metalearning-based manner is to construct a classifier with multiple learning episodes from the base dataset in the metatraining stage and use the trained classifier to perform metatesting in the novel dataset [9]. It should be noted that the categories in the novel dataset have not appeared in the base dataset for the few-shot learning problem.

Most few-shot remote sensing scene classification methods are based on metalearning, which can be summarized as the following aspects for improvement.

1) A pretraining stage with the based dataset is incorporated before the metatraining to obtain a more comprehensive feature extractor [10], [11], [12], [13], [14], [15], [16].
2) The attention mechanism [17] is designed on the backbone to extract more discriminative scene features [12], [13], [18], [19], [20], [21].
3) Novel metric methods or network architectures are proposed to construct a more discriminative metric space [18], [22], [23], [24], [25], [26], [27].
4) The semisupervised and self-supervised learning are integrated into metalearning to alleviate data scarcity [20], [21], [28].

Recently, another FSRSSC manner based on transfer learning has also attracted much attention [29]. In the transfer learning-based scheme, the feature extractor is trained with the base dataset first and then transferred for testing on the novel dataset. The feature extractor can either be further fine-tuned with the novel dataset in combination with a classifier or be directly fixed to extract features [30]. However, it usually leads to the collapse of the model since the categories in the novel dataset are not included in the base dataset. Consequently, it imposes strict requirements on the representation and generalization ability of the transferred model. Self-supervised learning, exploiting the structural information of the data itself by defining pretext tasks without involving manual labels, can easily develop supervisory signals to learn useful general representations [31], [32]. Therefore, self-supervised learning is usually designed as an auxiliary task

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to improve the generalization performance of the transfer few-shot scene classification model. Ji et al. [33] introduced two self-supervised pretext tasks of rotation and contrastive predictions. In [34], contrastive learning combined with a self-distillation strategy has been studied in multilayer features. Moreover, the practical self-supervised contrastive learning frameworks, Bootstrap Your Own Latent (BYOL) [35] and Simple Framework for Contrastive Learning of Visual Representations (SimCLR) [36], are also designed for metalearning in scene classification [16], [20], [21].

In previous works [16], [20], [33], self-supervised contrastive learning is mainly aimed at global features. However, in some extremely indistinguishable scenes, such as bridges and highways, their global features are difficult to be separated in the feature space. In human recognition, we can make judgmental inferences by connecting the subject of the scene with other targets. For example, rivers and land outside of the main objectives can be an important basis for identifying bridge and highway scenarios. Therefore, the correlations between local features in contrastive learning should also be considered. To alleviate the problem of high between-class similarity in few-shot scene classification, we propose a novel global–local contrastive learning auxiliary task. Specifically, global contrastive learning is able to learn a discriminative representation at the image level, while local contrastive learning further refines the contrastive features by exploring the connections between different objects in the scene. In terms of local contrastive learning, instead of directly performing contrastive learning on their local features, we pay more attention to the connection between the local features of the two augmented images. Specifically, we implement a pairwise self-attention operation on the local features of the two enhanced images, and two locally associated aligned images with self-attention weights can be obtained. The local similarity is calculated with the cosine similarity between the original and the aligned local features. Based on this, the global–local contrastive loss can be obtained with the global and local similarity based on the normalized temperature-scaled cross-entropy loss (also known as *NT-Xent*).

In addition to self-supervised contrastive learning, the other self-supervised auxiliary tasks also contribute greatly to the generalization. However, due to the complex relationship between different self-supervised auxiliary tasks, although the optimal combination of different pretext tasks may promote each other, it may also produce conflicts, resulting in overfitting or underfitting of the model. Self-distillation is a common solution for optimizing the complex model structure. In previous studies [37], [38], [39], only a single combination of self-supervision or self-distillation is designed. Chen et al. [40] designed an evolutionary multiobjective algorithm for self-supervised auxiliary tasks based on the heuristic search. While the Pareto self-supervised training [40] is able to obtain a tradeoff for multiple pretext tasks, it is also time-consuming in population evaluation. In addition, due to the different properties of remote sensing scene classification datasets, it is difficult to optimize uniformly with a single combination of self-supervised tasks. To simplify the complexity, we propose a multiform ensemble enhancement (MEE) algorithm in this article. Instead of designing sophisticated self-distillation strategies, we seek to train an ensemble of multiple feature extractors for generalization. For each classifier, we design a random combination of multiform self-supervised auxiliary tasks, integrating the few-shot main task loss and the global–local contrastive loss in training. With this design, we approach a single combination of self-supervised auxiliary tasks in algorithmic complexity but greatly reduce the time complexity compared with the traversal operation or evolutionary-based training strategies.

In conclusion, we propose a novel multiform ensemble self-supervised learning (MES^{2}L) framework for FSRSSC. To demonstrate the effectiveness of the proposed method, we first implement the MES^{2}L based on the transfer learning scheme, and the corresponding network is called MES^{2}L-Net. In MES^{2}L-Net, multiple classifiers are trained by three losses, i.e., the main task loss with the base dataset, global–local contrastive loss with the contrastive augmented data, and different forms of pretext tasks with self-supervised enhancement data. Then, we discard the fully connected layers of the classifiers, only freeze the feature extractors, and transfer them for testing. We employ the simple nearest class mean (NCM) and soft k-means (SKM) in testing to address the difficulties of data acquisition and data labeling in inductive and transductive settings, respectively. Moreover, the proposed MES^{2}L framework is designed in metalearning and metric learning to demonstrate its universality, while is not just limited to the transfer learning-based paradigm.

Our main contributions can be summarized as follows.

1) We propose a novel MES^{2}L framework for FSRSSC, and the MES^{2}L-Net is implemented in both inductive and transductive settings based on the transfer learning manner.
2) We propose a novel global–local contrastive learning auxiliary task to effectively solve the problem of low interclass separability. The self-attention mechanism (SAM) for local features can explore the interconnections between different objects in remote sensing scenes to obtain more discriminative feature representations.
3) We propose an MEE training method to obtain multiple backbones with randomly different combinations of self-supervised tasks. It not only approaches the best combined results of self-supervision and self-distillation in classification accuracy but also saves expensive computational resources to a great extent.
4) The proposed MES^{2}L-Net achieves state-of-the-art (SOTA) performance on three widely used remote sensing scene classification datasets. The combination with few-shot algorithms based on metalearning and metric learning also proves the effectiveness of the MES^{2}L framework in universality and generalization.

The remainder of this article is structured as follows. Section II briefly reviews the related work of few-shot scene classification, self-supervised few-shot learning, and the difference between inductive and transductive settings. In Section III, the proposed MES^{2}L-Net is described in detail. Section IV reports the experimental results of comparison

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and ablation. In addition, the generalization of the MES^{2}L framework is also reported in Section IV. Conclusions are drawn in Section V.

II. Related Work

A. Scene Classification With Few-Shot Learning

In few-shot learning, the base dataset and the novel dataset are used for training and testing, respectively. The base dataset usually contains a large amount of data for training a powerful feature extractor or a conditioned model, and the categories in the novel dataset are not encountered in the base dataset. According to the difference in training methods, it can be summarized in the following two manners in few-shot scene classification.

1) Based on Metalearning: Most of the scene classification few-shot learning algorithms are based on metalearning, i.e., episodic learning is used in the training phase [41], [42]. Taking N-way K-shot as an example, in each episode, N categories are randomly selected from the base dataset first, and then, K and Q images are randomly selected from these N categories as the support set and the query set, respectively. Based on this, Zhang et al. [10], [11], [12], Kim and Chi [13], Huang et al. [14], and Wang et al. [15] pretrain all categories on the base dataset and then transfer the trained feature extractor to metatraining, while PCFGNet [12], SAFFNet [13], AMN [18], DLA-MatchNet [19], SCL-MLNet [20], and task-specific contrastive (TSC) learning [21] explore the feature extractor in the metalearning framework with the attention mechanism for extracting more discriminative and abundant features. Furthermore, self-supervised learning is also combined in the metalearning framework in [16], [20], [21], and [28] as an auxiliary task to enhance the capability of the feature extractor. In terms of the improvement of metric space, RS-MetaNet [22] and RankingNet [25] are based on the relation network (RN) [43], in which [25] replaces the relation module with the relation learning subnetwork for knowledge distillation (KD). MKN [23] maps the low-dimensional to higher dimensional features for easier measurement decisions. In addition, the attention mechanism [18], [44] and calibration strategies [12], [24] to enhance the distinction between the support set and the query set are designed in IDLN [44], AMN [18], PCFGNet [12], and SPNet [24], respectively. References [26] and [27] refer to the framework of the transductive propagation network (TPN), including feature embedding, graph construction, label propagation, and loss calculation. The lifelong learning for scene recognition (LLSR) [45] is mainly based on the gradient iteration scheme for exploration.

2) Based on Transfer Learning: In this few-shot learning scheme, a feature extractor is first trained with the base dataset and then transferred to the testing phase with or without fine-tuning for the novel dataset. This scheme relies on large-scale data and powerful network architecture to obtain excellent generalization capability. In WRMatch [46], the random-weighting scheme of nuclear-norm and the self-weighting scheme are designed in semisupervised learning to ease the burden of data annotation. In DFM [47], the feature extractors for the support set and the query set are trained with the mini-ImageNet and tiered-ImageNet, respectively. The excellent results of DFM on remote sensing scene datasets demonstrate the plasticity of this transfer scheme when the amount of data is sufficiently abundant. Gong et al. [48] designed a quad-patch augmentation strategy and a two-path aggregation attention module to solve the few-shot scene classification from both data and architecture aspects. Furthermore, Ji et al. [33] and Zhou et al. [34] designed self-supervised auxiliary tasks or self-distillation strategies to further improve the generality of transfer.

B. Self-Supervised Few-Shot Learning

Few-shot learning has been widely explored in remote sensing fields [49], [50], such as image classification, object detection, and object segmentation. In the few-shot scene classification task, self-supervised learning is usually employed as an auxiliary task to train a more generalizable feature extractor for categories not seen in the downstream task [51]. In particular, contrastive learning is widely studied. Contrastive learning aims to learn an embedding space in which pairs of similar samples of the same class stay close to each other, while pairs of samples from different classes are far apart.

In SCL-MLNet [20], BYOL [35] is applied in the prototypical network (PN) [52] to enhance the ability of the feature extractor to learn more representative features from a small number of annotated samples. TSC [21] designed a supervised contrast learning method to enhance feature discrimination between intraclass and interclass scenes. Moreover, Gong et al. [16] designed SimCLR [36] in a metalearning framework to improve the applicability of the target domain, while Ji et al. [33] adopted a transfer-based few-shot scheme, employing two self-supervised pretext tasks, i.e., rotation and contrastive predictions. In [34], self-supervised contrastive learning combined with a self-distillation strategy has been studied in multilayer features. Overall, since the test classes are not encountered during training, it will lead to the collapse of the transfer model. Therefore, self-supervised auxiliary tasks are widely designed in transfer-based few-shot methods as a regularization term to enhance the generalization performance of the model.

C. Inductive and Transductive Settings in Few-Shot Learning

The difference between inductive and transductive learning is whether the entire query set has permission to be executed for the few-shot classifier. Fig. 1 illustrates an example of the difference between inductive and transductive settings in few-shot learning on the case of 3-way 1-shot. In the inductive setting, each sample is classified independently of the other, and other samples in the query set are available to the few-shot classifier. Therefore, a simple method is to employ the NCM classifier to measure the distance between the query sample to be classified and the support set [as shown in Fig. 1 (top)]. However, such an approach may lead to some misclassification, especially the samples that are difficult to classify among the various categories. As shown in Fig. 1 (top), the query sample is incorrectly classified as a blue category according to the NCM method. Therefore, other

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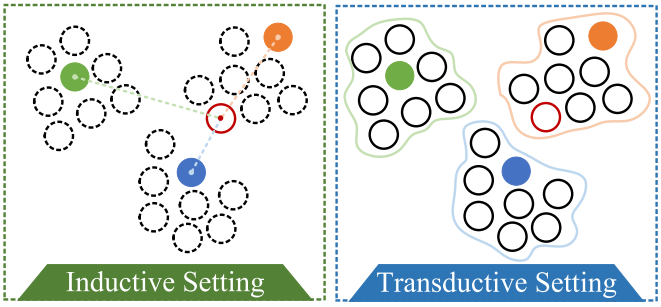


Fig. 1. **Illustration of the difference between inductive and transductive settings in few-shot learning.** Solid circles represent the labeled samples obtained from the support set, hollow circles are unlabeled samples from the entire query set, and the red hollow circle is the query sample to be classified. Solid circles of different colors stand for different categories. We show the case of 3-way 1-shot, and the number of query samples per class is set to 7.

few-shot metric-based algorithms have also been extensively studied using different distance metrics to obtain a better performance, such as the cosine distance in the Matching Network [53], the Euclidean distance in the PN [52], and the relation module in the RN [43].

On the contrary, in the transductive few-shot setting, both the support set and the query set can have access to the classifier [54]. Therefore, the query sample can be classified using the k-means clustering method combining the support set and the query set (as shown in Fig. 1 bottom). In addition, Liu et al. [55] proposed transductive reasoning to add a transduction mechanism, i.e., label propagation, in the framework of metalearning to deal with the problem of scarce data. The proposed TPN jointly learns the parameters of feature embedding and graph construction. Combining unsupervised learning and transductive learning to solve few-shot problems is studied in [56]. However, the computational cost of transductive learning is much more expensive than that of inductive learning; all inferences need to be overturned once the dataset is changed.

III. Methodology

A. Overview

We adopt the transfer-based scheme to solve the few-shot scene classification. Denote $\mathcal{D}^{\text{Train}}$, $\mathcal{D}^{\text{Valation}}$, and $\mathcal{D}^{\text{Test}}$ are the training, validation, and test sets, respectively. Different from the episode training strategy in metalearning-based methods, $\mathcal{D}^{\text{Train}}$ is consistent with traditional image classification and is not divided into the support set and the query set. The categories in the validation set and the test set are not included in the training set and are independent of each other. $\mathcal{D}^{\text{Valation}}$ and $\mathcal{D}^{\text{Test}}$ are divided into the support set and the query set, where the samples in the support set are labeled but the samples in the query set are not.

An overview of the proposed MES^{2}L-Net is shown in Fig. 2. The loss function for classifier training consists of two parts: one is the cross-entropy loss trained on the base dataset and the other is the self-supervised loss composed of the MEE and the global–local contrastive losses. Instead of designing complex distillation learning to alleviate the overfitting problem of few-shot networks, we directly train multiple classifiers with different forms to further enhance the generalization performance of the model. Subsequently, we discard the fully connected layers in the trained classifiers, only freeze all the feature extractors without further fine-tuning, and directly use them for testing. In the testing, we concatenate the feature vectors of the support set and the query set from multiple trained feature extractors. In addition, we provide two settings of inductive and transductive few-shot learning, which are correspondingly used to solve the two difficulties of data acquisition and labeling data in practical applications. Specifically, the feature vectors are preprocessed and projected onto the hypersphere, and then, classify the samples of the query set in the inductive and transductive settings with the NCM and SKM, respectively.

B. Global–Local Contrastive Learning

A difficult challenge for remote sensing scene classification is the large similarity between different scene categories, such as the indistinguishable scenes of bridges and highways, basketball courts, and tennis courts. The self-supervised contrastive learning aims to gather samples of the same category in adjacent regions in the feature space and widen the distance of dissimilar objects in the feature space. Therefore, the contrastive learning frameworks based on SimCLR [36] or BYOL [35] are explored in scene classification [16], [20], [33] to alleviate the above problem. However, the large similarity between classes leads to its limitation since the comparison is based on global features. Therefore, in addition to the global features, we also explore contrastive learning in local features to increase the contrast between high-similarity categories. The proposed global–local contrastive learning framework is illustrated in Fig. 3.

In terms of the global contrastive learning, the global feature is obtained with $z^g = \mathcal{P}_G(\mathcal{F}(\mathcal{I}))$, where $z^g \in \mathbb{R}^d$, $\mathcal{P}_G$, $\mathcal{F}$, and $\mathcal{I}$ are the global projection head, the encoder network, and the input image, respectively. For samples with batch size N_B, two independent data augmentations are first performed on each image. For a positive pair of examples (i, j), their global similarity can be calculated with cosine similarity, i.e.,

$$\text{sim}_G\left(z_i^g, z_j^g\right) = \frac{\left(z_i^g\right)^T z_j^g}{||z_i^g||\ ||z_j^g||}. \tag{1}$$

Therefore, the global contrastive learning loss function for (i, j) can be obtained as follows:

$$\ell_{\text{GC}}(i, j) = -\log \frac{\exp\left(\text{sim}_G\left(z_i^g, z_j^g\right)/\tau_g\right)}{\sum_{m=1}^{2N_B} \mathbb{1}_{[m\neq i]} \exp\left(\text{sim}_G\left(z_i^g, z_m^g\right)/\tau_g\right)} \tag{2}$$

where τ_g is the temperature parameter of global contrastive learning.

In terms of the local contrastive learning, we use the layer before the average pooling of the last layer as the local features and input it to the local projection head to obtain local contrast features, i.e., $z^l = \mathcal{P}_L(\mathcal{F}(\mathcal{I}))$, where $z^l \in \mathbb{R}^{h\times w\times d}$ and $\mathcal{P}_L$ is the local projection head. The features extracted from the convolutional layer closer to the output are the deep features of the model, which can extract some coarse-grained information. On the contrary, the shallow features

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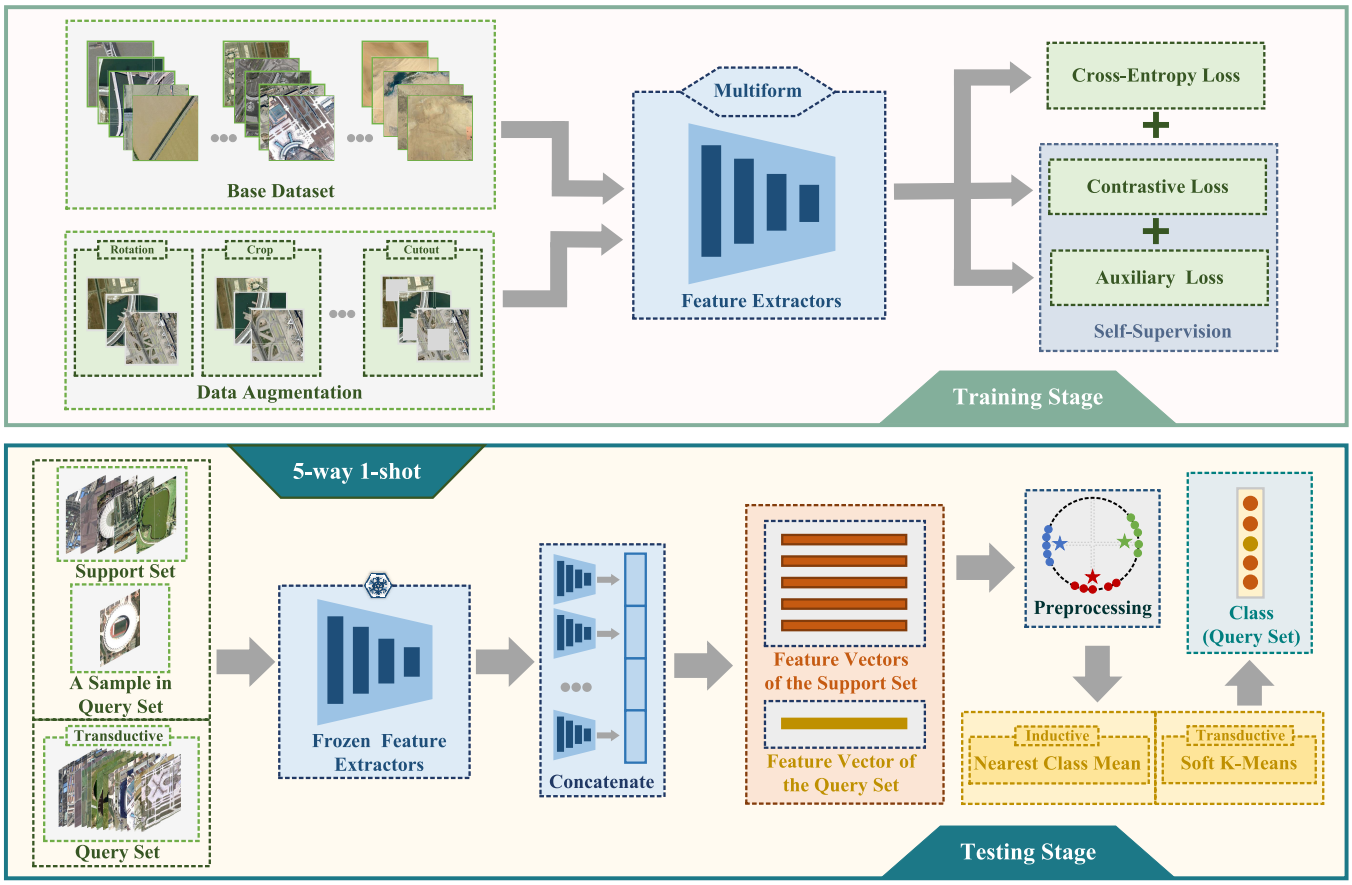


Fig. 2. **Overview.** In the training stage, we train multiple image classifiers (each classifier including a feature extractor and a fully connected layer) with three losses, i.e., cross-entropy loss with the base dataset, global–local contrastive loss with the contrastive augmented data, and auxiliary loss with multiform enhancement data. Then, we discard the last fully connected layers and freeze the feature extractors. In the testing stage, we obtained the features of the support set and the query set with multiple frozen pretrained feature extractors. For each image, multiple different feature vectors are output, and then, a concatenation operation is performed. Subsequently, the feature vectors are preprocessed. We employed the NCM and SKM to classify the sample in the query set in inductive and transductive settings, respectively. We show the case of a 5-way 1-shot in testing in the figure.

have higher resolution and contain more locations and detail information, but it is less semantic and more noisy due to fewer convolutions. Therefore, as the number of downsampling or convolution increases, the receptive field gradually increases, and the overlapping area between the receptive fields also increases. The information represented by the pixels at this time is the information of a region and the obtained region or phase feature information between neighboring regions, while the purpose of local contrastive learning is to employ the self-attention module to enhance the discrimination between scenes with high similarity, so it is more reasonable and effective to take advantage of the deep features.

Normally, the local similarity between two instances is directly compared with cosine similarity. However, it is hard to make an improvement because of the local features with high similarities, especially for those confusing categories. For some indistinguishable scenes, we usually combine other objects outside the target in the scene to assist the inference. Taking the scenes of the bridge and highway as an example, they have almost no difference in the characteristics of the road, but we can infer that the scene is a bridge based on the objects outside the road, such as rivers and lakes, while the highway can be inferred from the surrounding green belts and houses. Therefore, it inspires us to focus on the connection of different objects in indistinguishable scenes when exploring the contrast of local features.

Consequently, we designed a local contrastive learning strategy based on the self-attention module. In the SAM [57], there are three important variables, i.e., Query, Key, and Value. For a pair of examples (i, j), we take i as the anchor, the z_i^l is regarded as the Query and Value, and the Key is the z_j^l, and vice versa. The spatial dimension of local features is $h \times w \times d$, denoted z^s as the feature vector at the s spatial location, where $s \in [1, hw]$. For each z^s, ℓ_2 normalization is first performed. Then, for a pair of (z_i^l, z_j^l), the corresponding local similarity can be obtained as follows:

$$\begin{aligned}\mathrm{sim}_L\left(z_i^l, z_j^l\right) &= \frac{1}{hw}\sum_{s=1}^{hw}\left[\left(\left(z_i^l\right)^s\right)^T \mathrm{softmax}\left(\frac{\left(z_j^l\right)^s\left(\left(z_i^l\right)^s\right)^T}{\sqrt{d}}\right)\left(z_i^l\right)^s \right. \\ &\left. + \left(\left(z_j^l\right)^s\right)^T \mathrm{softmax}\left(\frac{\left(z_i^l\right)^s\left(\left(z_j^l\right)^s\right)^T}{\sqrt{d}}\right)\left(z_j^l\right)^s\right]. \end{aligned} \tag{3}$$

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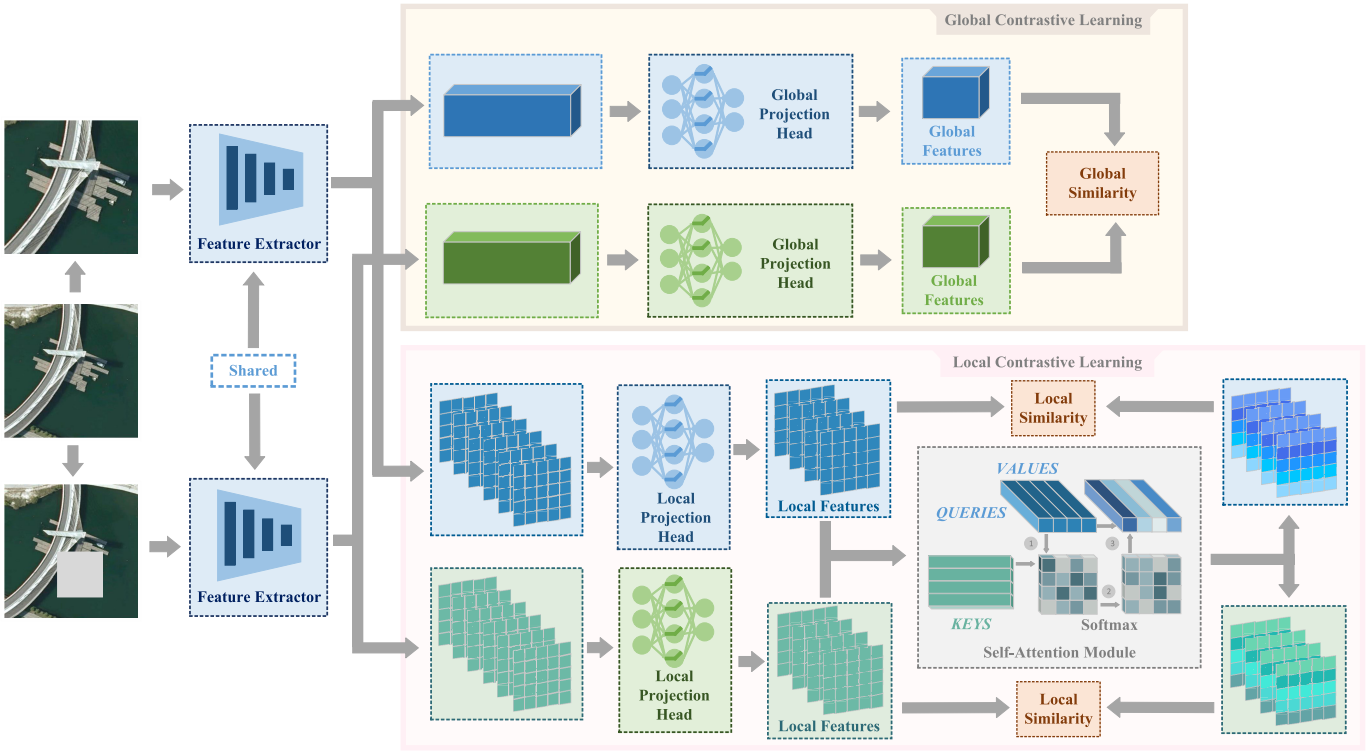


Fig. 3. **Global–local contrastive learning framework.** We use the features of the final output of the network and the output features before the average pooling layer of the last layer as the global and local features, respectively. The global and local projection heads are nonlinear multilayer perceptron (MLP) with one hidden layer. Taking the backbone of ResNet-12 as an example, the dimension of the resized input image is $\mathbb{R}^{84\times84\times3}$. After four residual convolution blocks, the dimension of the feature vector changes as $\mathbb{R}^{42\times42\times64}$, $\mathbb{R}^{21\times21\times160}$, $\mathbb{R}^{10\times10\times320}$, and $\mathbb{R}^{5\times5\times640}$. Then, the feature is processed by an average pooling layer of dimension $\mathbb{R}^{640}$. The dimension of local features is $\boldsymbol{z}^l = \mathcal{P}_L(\boldsymbol{h}^l)$, where $\boldsymbol{h}^l \in \mathbb{R}^{5\times5\times640}$ and $\boldsymbol{z}^l \in \mathbb{R}^{5\times5\times80}$. The dimension of global features is $\boldsymbol{z}^g = \mathcal{P}_G(\boldsymbol{h}^g)$, where $\boldsymbol{h}^g \in \mathbb{R}^{640}$ and $\boldsymbol{z}^g \in \mathbb{R}^{80}$.

Similar to (2), the local contrastive learning loss function can be obtained as follows:

$$\ell_{\mathrm{LC}}(i,j) = -\log \frac{\exp\left(\mathrm{sim}_L\left(\boldsymbol{z}_i^l, \boldsymbol{z}_j^l\right)/\tau_l\right)}{\sum_{m=1}^{2N_B} \mathbb{1}_{[m\neq i]} \exp\left(\mathrm{sim}_L\left(\boldsymbol{z}_i^l, \boldsymbol{z}_m^l\right)/\tau_l\right)} \tag{4}$$

where τ_l is the temperature parameter of local contrastive learning.

Overall, the loss function of global–local contrastive learning $\mathcal{L}_{\mathrm{CL}}$ can be defined as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{CL}} &= \mathcal{L}_{\mathrm{GC}} + \mathcal{L}_{\mathrm{LC}} \\ &= \frac{1}{2N_B}\sum_{m=1}^{2N_B}[\ell_{\mathrm{GC}}(2m-1,2m) + \ell_{\mathrm{GC}}(2m,2m-1) \\ &\qquad + \ell_{\mathrm{LC}}(2m-1,2m) + \ell_{\mathrm{LC}}(2m,2m-1)].\end{aligned} \tag{5}$$

C. Multiform Ensemble Enhancement

In the transfer-based few-shot learning scheme, self-supervised auxiliary tasks are currently receiving much attention. It is mainly used to improve the generalization ability of models under the scarcity of labeled data. Normally, the auxiliary tasks are optimized by minimizing the linear combination of task losses. However, it is difficult to decide which self-supervised auxiliary task is most effective for different datasets. The time cost will increase exponentially if the exhaustive method is used to conduct experiments. In addition, since the scenarios in the test set are not encountered during training, this imposes severe requirements on the generalization performance of the trained model.

To alleviate the above problems, we propose an MEE strategy to approximate the best combination of auxiliary tasks. We directly train multiple classifiers to alleviate model overfitting, instead of designing complex distillation methods [58], while the time required for MES^{2}L-Net is comparable to training a backbone with a combination of self-supervised tasks. In summary, the core of the MEE is to train multiple backbones with different self-supervised combinations. One is to replace KD to achieve model generalization, and the other is to shorten the time to pursue the optimal self-supervised combination. The pseudocode of the proposed MEE strategy is shown in Algorithm 1.

The core of MEE is to train multiple backbone networks through different random combinations of self-supervised auxiliary tasks to achieve model generalization. Suppose that there are X self-supervised auxiliary tasks candidates to use. In the training of each classifier, the loss function of multiform enhancement ($\mathcal{L}_{\mathrm{ME}}$) randomly selects one to four auxiliary tasks for equal combination (lines 3–5), where $\mathrm{Ramdom}_A(\Theta)$ refers to randomly selecting N from the set of self-supervised auxiliary tasks (Θ). Therefore, the bth classifier is trained with three main loss functions, i.e.,

$$\mathcal{L}^b = \mathcal{L}_{\mathrm{CE}} + \mathcal{L}_{\mathrm{CL}} + \mathcal{L}_{\mathrm{ME}}^b \tag{6}$$

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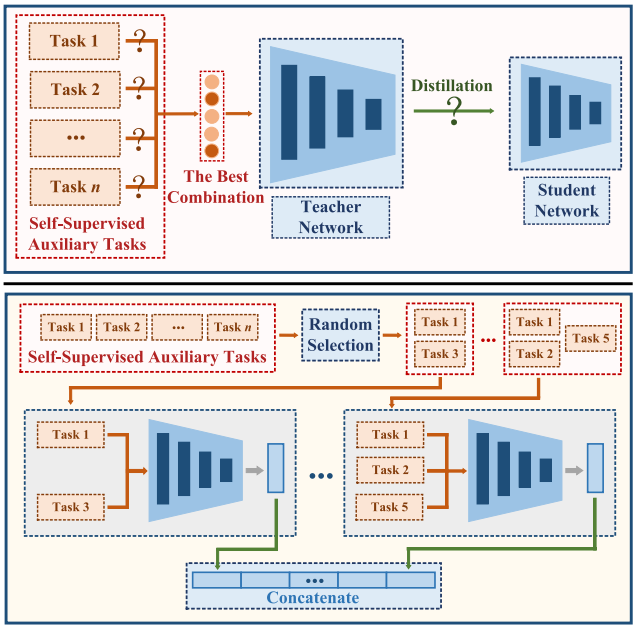


Fig. 4. Comparison of traversal operation for multiple self-supervised auxiliary tasks with KD and the proposed MEE.

Algorithm 1 MEE

Input: Θ: A set with X self-supervised auxiliary tasks, b_s: The number of backbones.

Output: b_s trained classifiers.

Set the initialization parameter $b \leftarrow 1$.
while $b \leq b_s$ **do**
  Generate a random integer A in the interval $[1, X]$.
  $\Theta_b \leftarrow Ramdom_A(\Theta)$.
  $\mathcal{L}^b_{ME} \leftarrow \sum_{a=1}^{A} \frac{1}{A}\mathcal{L}_a$,, where $\mathcal{L}_a \in E_b$.
  Train the b-th classifiers according to $\mathcal{L}^b = \mathcal{L}_{CE} + \mathcal{L}_{\mathrm{CL}} + \mathcal{L}^b_{ME}$.
  $b \leftarrow b + 1$.
end while

where $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy losses with the base dataset. $\mathcal{L}_{\mathrm{CL}}$ and $\mathcal{L}^b_{\mathrm{ME}}$ are the losses with the proposed global–local contrastive learning and multiform enhancement.

The improvement of the proposed MEE is illustrated in Fig. 4. Assuming that there are X pretext tasks and a distillation operation to be determined, we roughly formulate the training time cost of each task to be about Y, so the time complexity of the exhaustive method is about $\mathcal{O}((X+1)^2 Y)$, while the time complexity of the proposed MEE is $\mathcal{O}(XY)$. Moreover, it can not only greatly reduce the expensive time cost but also approach or exceed the best case in the exhaustive method in few-shot classification performance.

D. Testing

In practical applications, we may encounter difficulties in obtaining data or labeling data. Inductive learning is a better inference option when data are scarce, while transductive learning is superior when data labeling is difficult. In this article, we adopt both paradigms. Since the categories of the test set do not appear during training, the trained classifier is not suitable for testing. Therefore, we only keep the feature extractor $\mathcal{F}$ of the training set, and we can obtain the corresponding feature vector $\mathcal{Z} = \mathcal{F}(\mathcal{I})$ for any image $\mathcal{I}$. First, we need to preprocess the vector to center and project it onto the hypersphere [59] as follows:

$$\mathcal{Z}_{\mathrm{CH}} = \frac{\mathcal{Z} - \widetilde{\mathcal{Z}}}{||\mathcal{Z} - \widetilde{\mathcal{Z}}||_2} \tag{7}$$

where $\widetilde{\mathcal{Z}}$ is the average feature vector of the base dataset in inductive learning or the average feature vector of the number of few-shot problems in transductive learning. $||\cdot||_2$ is the ℓ_2 norm.

Denote the $\mathcal{S} = \{\mathcal{S}_1, \ldots, \mathcal{S}_k, \ldots, \mathcal{S}_N\}$ are the set of preprocessed feature vectors in the support set, where $\mathcal{S}_k$ is the set of feature vectors corresponding to the kth category. $\mathcal{Q}$ stands for the query set of preprocessed feature vectors.

1) Inductive Few-Shot Learning: We employed the NCM classifier [60] in the case of inductive few-shot learning. For each category, the class Barycenters from labeled samples are obtained as follows:

$$\widetilde{C}_k = \frac{1}{\mathcal{S}_k} \sum_{\mathcal{Z}_{\mathrm{CH}} \in \mathcal{S}_k} \mathcal{Z}_{\mathrm{CH}}. \tag{8}$$

Then, the samples in the query set can be predicted by the ℓ_2 distance with each Barycenter, i.e., $\forall \mathcal{Z}_{\mathrm{CH}} \in \mathcal{Q}$, we can obtain the corresponding category as follows:

$$\mathrm{Class}_I(\mathcal{Z}_{\mathrm{CH}}, [\widetilde{C}_1, \ldots, \widetilde{C}_N]) = \arg\min_k ||\mathcal{Z}_{\mathrm{CH}} - \widetilde{C}_N||_2. \tag{9}$$

2) Transductive Few-Shot Learning: We employed the SKM classifier [61] in the case of transductive few-shot learning. Considering that the Barycenters in transductive learning need to be recalculated each time, denoting the k as the sequence indexed, the initialization of $\widetilde{C}_k$ can be calculated as follows:

$$\widetilde{C}^0_k = \widetilde{C}_k \tag{10}$$

$$\widetilde{C}^{t+1}_k = \sum_{\mathcal{Z}_{\mathrm{CH}} \in \mathcal{S}_k \cup \mathcal{Q}} \frac{w\left(\mathcal{Z}_{\mathrm{CH}}, \widetilde{C}^t_k\right)}{\sum_{\mathcal{Z}'_{\mathrm{CH}} \in \mathcal{S}_k \cup \mathcal{Q}} w\left(\mathcal{Z}'_{\mathrm{CH}}, \widetilde{C}^t_k\right)} \mathcal{Z}_{\mathrm{CH}} \tag{11}$$

where $w(\mathcal{Z}_{\mathrm{CH}}, \widetilde{C}^t_k)$ is a weighting function on $\mathcal{Z}_{\mathrm{CH}}$ that provides the probability corresponding to the center of gravity $\widetilde{C}_k$, and the weighted value is calculated by the decreasing function of ℓ_2 distance between the data points and the class Barycenters, which can be defined as follows:

$$w\left(\mathcal{Z}_{\mathrm{CH}}, \widetilde{C}^t_k\right) = \begin{cases} 1, & \text{if } \mathcal{Z}_{\mathrm{CH}} \in \mathcal{S}_i \\ \dfrac{\exp\left(-\alpha \left\| \mathcal{Z}_{\mathrm{CH}} - \widetilde{C}^t_k \right\|_2^2\right)}{\sum_{k=1}^{N} \exp\left(-\alpha \left\| \mathcal{Z}_{\mathrm{CH}} - \widetilde{C}^t_k \right\|_2^2\right)}, & \text{if } \mathcal{Z}_{\mathrm{CH}} \in \mathcal{Q} \end{cases}$$

where α is the temperature value. In this article, we set $\alpha = 5$ as in [59] to be consistent across datasets and backbones. In addition, we use a limited number of steps to compute (11)

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to obtain the final result vector $\widetilde{C}_k^{\infty}$. Therefore, for all samples $\mathcal{Z}_{\mathrm{CH}}$ in the query set, it can be classified as follows:

$$\mathrm{Class}_T\left(\mathcal{Z}_{\mathrm{CH}}, \left[\widetilde{C}_1^{\infty} \ \ldots, \widetilde{C}_N^{\infty}\right]\right) = \arg\min_k \left\| \mathcal{Z}_{\mathrm{CH}} - \widetilde{C}_N^{\infty} \right\|_2. \tag{12}$$

IV. Experimental Results

A. Datasets

In this article, three benchmark remote sensing scene classification datasets are employed to demonstrate the effectiveness of our proposed method, i.e., the UC Merced_LandUse dataset [62], the WHU-RS19 dataset [63], and the NWPU-RESISC45 dataset [64]. To achieve a fair comparison, we follow the slicing method of training, validation, and test sets in [14], [19], [24], and [26] for these three datasets. In addition, to be consistent, all the scene images in these three datasets are resized to 84 × 84 pixels.

1) UC Merced_LandUse Dataset [62]: The UC Merced Computer Vision Laboratory published the UC Merced_LandUse dataset containing 2100 scene images in 21 categories. The scene images are obtained from the United States Geological Survey (USGS) National Map of 20 regions, with a spatial resolution of 0.3 m and a size of 256 × 256 pixels. In the experiments, consistent with previous few-shot scene classification works, the training, validation, and test sets have ten, six, and five classes, respectively.

2) WHU-RS19 Dataset [63]: Wuhan University released the WHU-RS19 dataset containing 1005 scene images in 19 categories in 2012. There are at least 50 scene images in each category with a size of 600 × 600 pixels. Variations in lighting, scale, resolution, and viewpoint-dependent appearance in some categories make this dataset more complicated. For this dataset, nine, five, and five classes are divided into training, validation, and test sets.

3) NWPU-RESISC45 Dataset [64]: The NWPU-RESISC45 dataset was released by Northwestern Polytechnical University in 2017, which is mainly collected in more than 100 countries and regions from Google Earth. This dataset has a total of 45 scene categories, and each category contains 700 scene images with 256 × 256 pixels. Except for a few scene images with low spatial resolution, such as islands, lakes, and mountains, most of the images have a spatial resolution of 30–0.2 m. In the experiments, 25, ten, and ten classes are used as training, validation, and test sets, respectively.

B. Implementation Details

To ensure the fairness and comparability of experiments, we implemented all the experiments on the Pytorch framework with CUDA 11.6 by using a single NVIDIA RTX 3090. To eliminate the expensive time cost of training multiple backbones, we employed the snapshot ensembles training strategy [65]. The data augmentation strategy in contrastive learning is consistent with SimCLR [36]. We train the model for 500 epochs with a batch size of 16. The learning rate is set to 0.05. In addition, we employed the stochastic gradient descent (SGD) optimizer with a momentum of 0.9. Consistent with conventional few-shot studies, the number of samples per class in the query set is set to 15. The overall accuracy of 5-way 1-shot and 5-way 5-shot with 95% confidence intervals is reported. The source code is available at https://github.com/OMEGAFSL/MESSL.

In terms of the self-supervised auxiliary tasks, according to the prior knowledge of previous studies, four commonly used auxiliary tasks, including rotation prediction ($\mathcal{L}_r$) [33], [37], [66], patch location ($\mathcal{L}_l$) [37], [67], jigsaw puzzle ($\mathcal{L}_p$) [38], [68], and manifold mixup ($\mathcal{L}_m$) [69], are employed in the self-supervised pretext task set in this article.

C. Comparison With Other Methods

We implement the MES^{2}L-Net in both inductive and transductive settings based on the ResNet-12 backbone. The vast majority of comparison algorithms are based on metalearning, including the optimization-based: MAML [70], Meta-SGD [71], and LLSR [45], and the metric learning-based: MatchingNet (MN) [53], PN [52], RN [43], TPN [55], RS-MetaNet [22], DLA-MatchNet [19], PCFGNet [12], IDLN [44], MKN [23], GES-Net [26], SCL-MLNet [20], SPNet [24], TAE-Net [14], and TSC [21], while *baseline* [72], S2M2 [69], Ji et al. [33], our previous work DANet [48], and the proposed MES^{2}L-Net are based on the transfer learning-based few-shot scheme. In addition to TPN and GES-Net, which are based on transductive settings, all other peer competitors are based on inductive settings during testing. The comparative experimental results are reported from their respective papers. The best and the second best results are shown in bold and underlined, respectively.

Tables I–III show the comparative experimental results on UC Merced, WHU-RS19, and NWPU-RESISC45 datasets, respectively. It can be observed that our proposed MES^{2}L-Net achieves SOTA results in both inductive and transductive settings. As summarized in [72], training a feature extractor with excellent generalization performance is usually more effective than designing sophisticated metalearning-based algorithms, which is attributed to the fact that the transfer learning-based few-shot algorithms fully explore the feature relationship between the training samples.

On the UC Merced dataset, MES^{2}L-Net is 2.61% and 2.93% higher than the previous SOTA in the inductive setting (i.e., DANet) on 1-shot and 5-shot, respectively. DANet mainly makes improvements in terms of data augmentation and network architecture, while MES^{2}L-Net focuses on employing the self-supervised auxiliary tasks to increase pseudo data to improve the generalization performance of the feature extractor. In the transductive setting, MES^{2}L-Net outperforms the previous best algorithm, i.e., GES-Net, by 11.85% and 1.26% in 1-shot and 5-shot, respectively. The excellent improvement on 1-shot is mainly due to the excellent discriminability of MES^{2}L-Net from the global–local contrastive learning.

The improvement of MES^{2}L-Net on the WHU-RS19 dataset with a small amount of data is also significant. Specifically, the method proposed by Ji et al. [33] employed the self-supervised rotation and contrastive auxiliary task and achieved very good performance in the inductive setting. While our proposed

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TABLE I

OVERALL ACCURACY (IN %) COMPARISON OF DIFFERENT FEW-SHOT LEARNING METHODS ON THE UC MERCED DATASET, WHERE THE "†" INDICATES TESTING WITH TRANSDUCTIVE SETTING

Method	5-way	
	1-shot	5-shot
%Optimization-based		
MAML [70]	48.86 ± 0.74	60.78 ± 0.62
Meta-SGD [71]	50.52 ± 2.61	60.82 ± 2.00
LLSR [45]	39.47	57.40
%Metric-based		
MatchingNet [53]	46.46 ± 0.75	67.39 ± 0.50
Prototypical Net [52]	54.03 ± 0.78	70.81 ± 0.48
Relation Net [43]	49.77 ± 0.78	57.98 ± 0.57
TPN† [55]	53.36 ± 0.77	68.23 ± 0.52
RS-MetaNet [22]	57.23 ± 0.56	76.08 ± 0.28
DLA-MatchNet [19]	53.76 ± 0.62	63.01 ± 0.51
MKN [23]	58.45 ± 0.54	77.92 ± 0.32
SCL-MLNet [20]	51.37 ± 0.79	68.09 ± 0.92
GES-Net† [26]	58.88 ± 0.81	$\mathbf{81.66 \pm 0.50}$
TAE-Net [14]	60.21 ± 0.72	77.44 ± 0.51
SPNet [24]	57.64 ± 0.73	73.52 ± 0.51
TSC [21]	55.11 ± 0.68	69.20 ± 0.64
%Transfer-based		
baseline [72]	60.41 ± 0.38	77.02 ± 0.24
S2M2 [69]	56.42 ± 0.40	71.97 ± 0.27
DANet [48]	61.60 ± 0.18	78.02 ± 0.16
MES^{2}L-Net (Ours)	$\mathbf{64.21 \pm 0.17}$	80.95 ± 0.15
MES^{2}L-Net† (Ours)	$\mathbf{70.73 \pm 0.16}$	$\mathbf{82.92 \pm 0.12}$

TABLE II

OVERALL ACCURACY (IN %) COMPARISON OF DIFFERENT FEW-SHOT LEARNING METHODS ON THE WHU-RS19 DATASET, WHERE THE "†" INDICATES TESTING WITH TRANSDUCTIVE SETTING

Method	5-way	
	1-shot	5-shot
%Optimization-based		
MAML [70]	49.13 ± 0.65	62.49 ± 0.51
Meta-SGD [71]	51.54 ± 2.31	61.74 ± 2.02
LLSR [45]	57.10	70.65
%Metric-based		
MatchingNet [53]	60.14 ± 0.67	68.66 ± 0.38
Prototypical Net [52]	69.03 ± 0.65	84.84 ± 0.33
Relation Net [43]	70.01 ± 0.68	86.04 ± 0.32
TPN† [55]	66.51 ± 0.87	78.50 ± 0.56
DLA-MatchNet [19]	68.27 ± 1.83	79.89 ± 0.33
IDLN [44]	73.89 ± 0.88	83.12 ± 0.56
PCFGNet [12]	72.41 ± 0.91	85.26 ± 0.66
GES-Net† [26]	75.84 ± 0.78	82.37 ± 0.38
TAE-Net [14]	73.67 ± 0.74	88.95 ± 0.52
SPNet [24]	81.06 ± 0.60	88.04 ± 0.28
TSC [21]	70.99 ± 0.74	82.18 ± 0.32
%Transfer-based		
baseline [72]	75.57 ± 0.36	88.65 ± 0.18
S2M2 [69]	69.00 ± 0.41	82.14 ± 0.21
Ji *et al.* [33]	86.29 ± 0.30	92.96 ± 0.12
DANet [48]	75.02 ± 0.16	89.21 ± 0.07
MES^{2}L-Net (Ours)	$\mathbf{87.21 \pm 0.12}$	$\mathbf{93.58 \pm 0.07}$
MES^{2}L-Net†(Ours)	$\mathbf{92.21 \pm 0.05}$	$\mathbf{95.03 \pm 0.04}$

MES^{2}L-Net mainly designs a novel global–local contrastive learning loss, the MEE training strategy can take advantage of multiple self-supervised auxiliary tasks to obtain a feature extractor with stronger generalization performance. In the inductive setting, MES^{2}L-Net increased by 12.19% and 4.37% compared with DANet on 1-shot and 5-shot, respectively. It is particularly noteworthy that the accuracy of MES^{2}L-Net on 1-shot is even more than 95% in the transductive setting, which is due to the great improvement of discriminant in only a small number of samples by the design of global–local contrastive learning.

TABLE III

OVERALL ACCURACY (IN %) COMPARISON OF DIFFERENT FEW-SHOT LEARNING METHODS ON THE NWPU-RESISC45 DATASET, WHERE THE "†" INDICATES TESTING WITH TRANSDUCTIVE SETTING

Method	5-way	
	1-shot	5-shot
%Optimization-based		
MAML [70]	48.40 ± 0.82	62.90 ± 0.69
Meta-SGD [71]	60.63 ± 0.90	75.75 ± 0.65
LLSR [45]	51.43	72.90
%Metric-based		
MatchingNet [53]	45.80 ± 0.86	64.35 ± 0.65
Prototypical Net [52]	62.43 ± 0.84	78.68 ± 0.55
Relation Net [43]	66.58 ± 0.91	76.11 ± 0.58
TPN† [55]	66.51 ± 0.87	78.50 ± 0.56
RS-MetaNet [22]	52.78 ± 0.09	71.49 ± 0.81
DLA-MatchNet [19]	68.80 ± 0.70	81.63 ± 0.46
IDLN [44]	75.25 ± 0.75	84.67 ± 0.23
PCFGNet [12]	72.05 ± 0.75	85.07 ± 0.45
GES-Net† [26]	70.83 ± 0.85	82.27 ± 0.55
MKN [23]	65.84 ± 0.89	82.67 ± 0.55
SCL-MLNet [20]	62.21 ± 1.12	80.86 ± 0.76
TAE-Net [14]	69.13 ± 0.83	82.37 ± 0.52
SPNet [24]	67.84 ± 0.87	83.94 ± 0.50
TSC [21]	73.26 ± 0.15	84.62 ± 0.35
%Transfer-based		
baseline [72]	69.02 ± 0.46	85.62 ± 0.25
S2M2 [69]	63.24 ± 0.47	83.23 ± 0.28
Ji *et al.* [33]	76.70 ± 0.44	89.87 ± 0.21
DANet [48]	74.30 ± 0.20	87.29 ± 0.11
MES^{2}L-Net (Ours)	$\mathbf{79.37 \pm 0.20}$	$\mathbf{89.94 \pm 0.11}$
MES^{2}L-Net† (Ours)	$\mathbf{86.55 \pm 0.18}$	$\mathbf{91.06 \pm 0.11}$

In terms of the NWPU-RESISC45 dataset, MES^{2}L-Net performs well on both 1-shot and 5-shot. The metric learning-based algorithms SPNet and IDLN also achieved very good results, which are mainly designed on metric space and network architecture, respectively. For the proposed MES^{2}L-Net, in the metric space, it directly employed the NCM and SKM to measure the feature vectors between the support set and the query set in inductive and transductive settings, respectively. In addition, there is no spatial or channel attention module introduced in the network architecture in MES^{2}L-Net. Therefore, MES^{2}L-Net with higher accuracy is easier to deploy and implement than other complex metalearning-based algorithms.

D. Ablation Experiments

1) General Ablation: Similarly, we report the results of all ablation experiments on the ResNet-12 backbone with both inductive and transductive settings, where the "Baseline" refers to only one backbone trained with the base dataset based on

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TABLE IV
EXPERIMENT RESULTS (IN %) OF THE GENERAL ABLATION ON THREE REMOTE SENSING SCENE CLASSIFICATION DATASETS

Method	UC Merced		WHU-RS19		NWPU-RESISC45	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Inductive Setting						
Baseline	50.99 ± 0.24	69.36 ± 0.17	65.36 ± 0.16	82.74 ± 0.09	65.64 ± 0.21	80.65 ± 0.13
+ Global-Local Contrastive Learning	59.74 ± 0.18	75.12 ± 0.16	79.14 ± 0.15	88.86 ± 0.08	74.91 ± 0.21	86.58 ± 0.15
+ Multiform Ensemble Enhancement	58.63 ± 0.18	73.14 ± 0.16	83.20 ± 0.13	91.43 ± 0.07	74.60 ± 0.20	85.47 ± 0.14
+ Both	$\mathbf{64.21 \pm 0.17}$	$\mathbf{80.95 \pm 0.15}$	$\mathbf{87.21 \pm 0.12}$	$\mathbf{93.58 \pm 0.07}$	$\mathbf{79.37 \pm 0.20}$	$\mathbf{89.94 \pm 0.11}$
Transductive Setting						
Baseline	52.98 ± 0.24	74.01 ± 0.15	70.16 ± 0.21	84.30 ± 0.10	73.07 ± 0.29	81.90 ± 0.15
+ Global-Local Contrastive Learning	64.39 ± 0.19	79.40 ± 0.13	86.76 ± 0.17	91.83 ± 0.06	82.99 ± 0.21	87.48 ± 0.13
+ Multiform Ensemble Enhancement	62.46 ± 0.20	76.29 ± 0.14	88.80 ± 0.17	92.32 ± 0.06	81.63 ± 0.21	86.60 ± 0.14
+ Both	$\mathbf{70.73 \pm 0.16}$	$\mathbf{82.92 \pm 0.12}$	$\mathbf{92.21 \pm 0.05}$	$\mathbf{95.03 \pm 0.04}$	$\mathbf{86.55 \pm 0.18}$	$\mathbf{91.06 \pm 0.11}$

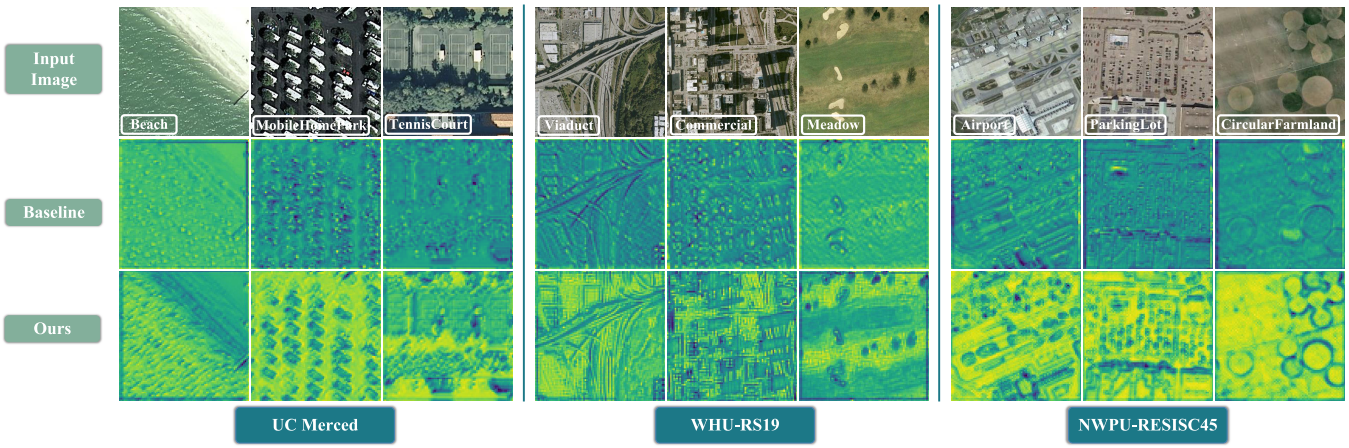


Fig. 5. **Comparison of shallow feature maps between "Baseline" and our proposed MES^{2}L-Net.** Here, we show the feature maps generated by the first convolutional layer. Compared with "Baseline," models generated by MES^{2}L-Net have clearer outlines of key scene targets, which provides higher discriminativeness of scene features for subsequent few-shot classification.

the cross-entropy loss. Table IV presents the general ablation results for the two innovations, i.e., the global–local contrastive learning and the MEE. The experimental results show that both the proposed components can improve the classification performance, and the combination of the two can achieve the best improvement. It can be summarized as the importance of self-supervised auxiliary tasks for model generalization performance, and multibackbone ensemble training is an effective alternative to distillation. In addition, Fig. 5 also shows the improvement results of our proposed MES^{2}L-Net in shallow feature map visualization. It can be observed that our proposed method has a stronger feature extraction ability for objectives in different scenes, which provides a basic guarantee for the downstream few-shot classification.

2) Ablation on the Global–Local Contrastive Learning: In this ablation experiment, we focus on demonstrating the effectiveness of global and local contrastive learning (GLCL). The detailed global–local contrastive learning ablation results are shown in Table V, where "+GCL" refers to employing only global contrastive loss based on SimCLR [36]. "+LCL w/o SAM" means that only contrastive learning is used in local features, and the SAM is not integrated, while "+LCL w/ SAM" is designed with the SAM.

The experimental results show that both the introduction of GLCL loss can improve the classification accuracy. Therefore, consistent with the experimental results in [33], self-supervised contrastive learning can improve the model generalization of the transfer learning-based few-shot framework. Compared with only using the global features, the addition of the local features with SAM can improve the UCM Merced, WHU-RS19, and NWPU-RESISC45 datasets in the inductive setting by 4.38%, 5.26%, and 6.52% in 5-way 1-shot, and 2.45%, 4.06%, and 2.86% in 5-way 5-shot, respectively. In the transductive setting, 7.11%, 10.00%, and 5.45% in 5-way 1-shot, and 4.11%, 4.51%, and 3.43% in 5-way 5-shot are boosted in three corresponding remote sensing datasets. However, if we only simply perform the contrastive learning on local features, the improvement effect is not obvious, which is also the original intention of designing a novel local contrastive learning loss with the SAM. With our design for contrastive learning of local features, the SAM is able to explore the connections between different objects in the scenes, thereby further separating different categories of scenes in the feature space.

The partitioning method of the dataset in [14], [19], [24], and [26] is not particularly low in between-class separability in the validation and test sets. Therefore, to further demonstrate the effectiveness of global–local contrastive learning in addressing the problem of low interclass separability in few-shot scene classification, we reclassify the

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TABLE V

ABLATION RESULTS (IN %) OF THE GLOBAL–LOCAL CONTRASTIVE LEARNING ON THREE REMOTE SENSING SCENE CLASSIFICATION DATASETS

Method	UC Merced		WHU-RS19		NWPU-RESISC45	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Inductive Setting						
Baseline	50.99 ± 0.24	69.36 ± 0.17	65.36 ± 0.16	82.74 ± 0.09	65.64 ± 0.21	80.65 ± 0.13
+ GCL [36]	55.36 ± 0.20	72.67 ± 0.13	73.88 ± 0.17	84.80 ± 0.10	68.39 ± 0.20	83.72 ± 0.14
+ LCL w/o SAM	53.62 ± 0.21	71.28 ± 0.13	72.00 ± 0.18	84.09 ± 0.10	67.64 ± 0.21	82.39 ± 0.15
+ LCL w/ SAM	55.83 ± 0.20	73.14 ± 0.12	75.02 ± 0.17	85.28 ± 0.06	69.54 ± 0.20	83.98 ± 0.15
+ Both w/o $\text{LCL}_{+\text{SAM}}$	58.11 ± 0.18	74.05 ± 0.12	76.96 ± 0.14	86.97 ± 0.05	72.09 ± 0.00	85.47 ± 0.14
+ Both w/ $\text{LCL}_{+\text{SAM}}$	$\mathbf{59.74 \pm 0.18}$	$\mathbf{75.12 \pm 0.16}$	$\mathbf{79.14 \pm 0.15}$	$\mathbf{88.86 \pm 0.08}$	$\mathbf{74.91 \pm 0.21}$	$\mathbf{86.58 \pm 0.15}$
Transductive Setting						
Baseline	52.98 ± 0.24	74.01 ± 0.15	70.16 ± 0.21	84.30 ± 0.10	73.07 ± 0.29	81.90 ± 0.15
+ GCL [36]	57.28 ± 0.21	75.29 ± 0.14	76.76 ± 0.21	87.32 ± 0.08	77.54 ± 0.20	84.05 ± 0.13
+ LCL w/o SAM	56.38 ± 0.21	74.87 ± 0.14	74.06 ± 0.22	86.71 ± 0.09	75.00 ± 0.21	83.62 ± 0.15
+ LCL w/ SAM	57.32 ± 0.21	75.88 ± 0.14	78.14 ± 0.21	87.72 ± 0.08	78.60 ± 0.20	84.77 ± 0.15
+ Both w/o $\text{LCL}_{+\text{SAM}}$	62.15 ± 0.20	78.02 ± 0.13	84.64 ± 0.17	89.39 ± 0.06	80.09 ± 0.20	86.58 ± 0.15
+ Both w/ $\text{LCL}_{+\text{SAM}}$	$\mathbf{64.39 \pm 0.19}$	$\mathbf{79.40 \pm 0.13}$	$\mathbf{86.76 \pm 0.17}$	$\mathbf{91.83 \pm 0.06}$	$\mathbf{82.99 \pm 0.21}$	$\mathbf{87.48 \pm 0.13}$

TABLE VI

SPLIT DETAILS ON THE NWPU-RESISC45 DATASET WITH LOW INTER-CLASS SEPARABILITY IN THE VALIDATION AND TEST SETS

Train	Airplane; Airport; Beach; Chaparral; Church; Cloud; Circular Farmland; Desert; Harbor; Intersection; Island; Lake; Meadow; Mountain; Palace; Railway Station; Railway; River; Rectangular Farmland; Sea Ice; Ship; Snowberg; Storage Tank; Terrace; Thermal Power Station
Validation	Commercial Area; Industrial Area; Dense Residential; Medium Residential; Sparse Residential; Parking Lot; Mobile Home Park; Golf Course; Forest; Wetland
Test	Baseball Diamond; Basketball Court; Ground Track Field; Stadium; Tennis Court; Bridge; Freeway; Overpass; Runway; Roundabout

TABLE VII

ABLATION RESULTS (IN %) OF THE GLOBAL–LOCAL CONTRASTIVE LEARNING ON THE REPARTITIONED NWPU-RESISC45 DATASET (CORRESPONDING TO TABLE VI)

Method	NWPU-RESISC45	
	1-shot	5-shot
Inductive Setting		
Baseline	53.94 ± 0.20	69.55 ± 0.16
+ GCL [36]	58.45 ± 0.20	72.79 ± 0.15
+ LCL w/o SAM	56.61 ± 0.20	71.20 ± 0.15
+ LCL w/ SAM	58.48 ± 0.20	74.60 ± 0.15
+ GLCL_{MoCo} [73]	60.61 ± 0.22	71.59 ± 0.16
+ GLCL_{BYOL} [35]	63.32 ± 0.21	74.91 ± 0.16
+ $\text{GLCL}_{SimSiam}$ [74]	64.25 ± 0.19	77.03 ± 0.14
+ Both w/o $\text{LCL}_{+\text{SAM}}$ [36]	64.03 ± 0.20	76.41 ± 0.15
+ Both w/ $\text{LCL}_{+\text{SAM}}$	$\mathbf{66.38 \pm 0.20}$	$\mathbf{77.14 \pm 0.15}$
Transductive Setting		
Baseline	61.97 ± 0.29	70.63 ± 0.18
+ GCL [36]	64.16 ± 0.23	74.41 ± 0.16
+ LCL w/o SAM	63.40 ± 0.23	73.77 ± 0.16
+ LCL w/ SAM	65.85 ± 0.23	75.81 ± 0.16
+ GLCL_{MoCo} [73]	64.24 ± 0.23	72.50 ± 0.18
+ GLCL_{BYOL} [35]	69.78 ± 0.22	75.38 ± 0.17
+ $\text{GLCL}_{SimSiam}$ [74]	70.47 ± 0.22	78.94 ± 0.15
+ Both w/o $\text{LCL}_{+\text{SAM}}$ [36]	70.07 ± 0.22	78.02 ± 0.15
+ Both w/ $\text{LCL}_{+\text{SAM}}$	$\mathbf{71.38 \pm 0.22}$	$\mathbf{81.85 \pm 0.15}$

indistinguishable classes into validation and test sets on the NWPU-RESISC45 dataset, as shown in Table VI. In addition, to demonstrate the effectiveness of our design of GLCL with SAM, we also compare the methods of four GLCL frameworks, i.e., MoCo [73], SimCLR [36], BYOL [35], and SimSiam [74]. It is worth noting that "+Both w/o LCL_{+}SAM" is the result of GLCL based on the SimCLR [36].

The experimental results in Table VII show that the design of the SAM in local contrastive learning is superior in solving the problem of high interclass similarity. In the case of local contrastive learning without the local SAM, 2.67% and 1.65% in the inductive setting are improved in 5-way 1-shot and 5-way 5-shot, respectively. 1.43% and 3.14% in the transductive setting are improved in 5-way 1-shot and 5-way 5-shot, respectively. However, after adding the self-attention mechanism in local contrastive learning, 4.54% and 5.05% in the inductive setting are improved in 5-way 1-shot and 5-way 5-shot, respectively. 3.88% and 5.18% in the transductive setting are improved in 5-way 1-shot and 5-way 5-shot, respectively. Combined with global contrastive learning, the same improvement can still be verified. Fig. 6 further demonstrates the rationality of our design of global–local contrastive learning from the comparison of heat map visualization. In terms of comparison with other GLCL frameworks, the GLCL based on the SimCLR is not as good as the results based on SimSiam. However, after adding an SAM to the local features, the experimental results of the inductive and transductive settings can reach the optimum. Some failure cases are also given in Fig. 7. MES^{2}L-Net also encounters failures when the surrounding objects of the scene subject are similar, especially in the 1-shot case.

3) Ablation on the Multiform Ensemble Enhancement: Fig. 8 shows the impact of multiple backbones on classification performance relative to the "Baseline," i.e., only employing the cross-entropy loss for training. The experimental results show that the connection of multiple backbone networks for training has better classification accuracy than that of only a single

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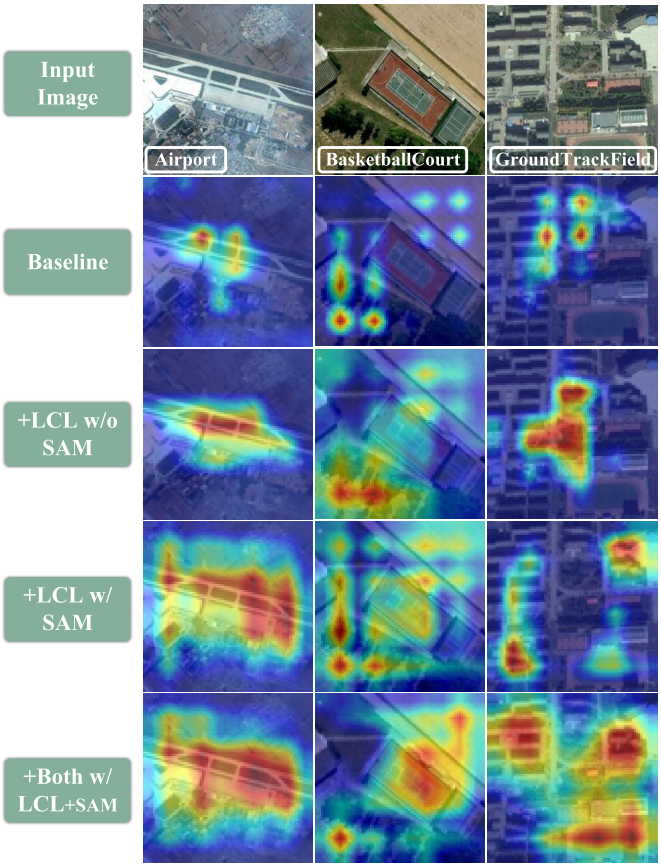


Fig. 6. **Comparison of heat map visualization between "Baseline" and our proposed MES^{2}L-Net.** The proposed global–local contrastive learning can make the model have a larger receptive field and focus not only on the main objects of the scene but also on the local key objects.



Fig. 7. Some failure cases of MES^{2}L-Net in the global–local contrastive learning ablation experiment.

backbone network in both inductive and transductive settings. However, it does not mean that the greater the number of network backbones, the better the generalization performance of the network. It has no effect on the results or even reduces the classification accuracy when there are more than four backbones.

Compared to the exhaustive method of the best results in different numbers of self-supervised auxiliary tasks, Fig. 9 shows the experimental results of the proposed MEE training on three remote sensing datasets in both inductive and transductive settings, where the "SSL" refers to the best combination of self-supervised tasks on a fixed number of pretext tasks, and

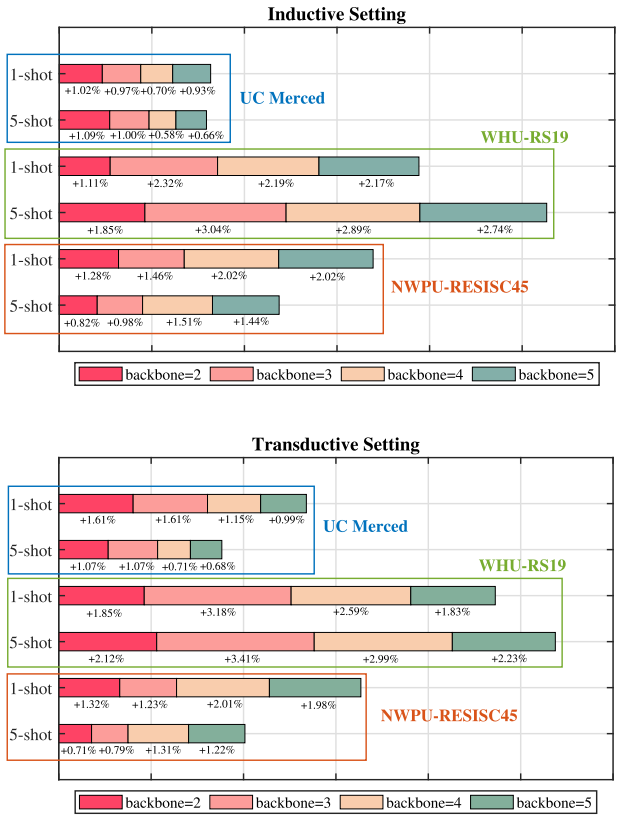


Fig. 8. Performance improvement for different numbers of backbones relative to the "Baseline".

TABLE VIII
COMPARISON OF THE MEE AND THE EXHAUSTIVE METHOD IN TIME (GPU DAYS)

Datasets	MES^{2}L-Net	Traversal Operation
UC Merced	0.0232	0.2894
WHU-RS19	0.0289	0.3357
NWPU-RESISC45	0.5151	5.5267

"SSL + KD" denotes that KD is added on the basis of "SSL." The experimental results show that the self-distillation can improve the generalization of the network to obtain higher classification accuracy in most cases. In addition, it is not that the combination of more self-supervised auxiliary tasks can improve the model. This is because, when multiple pretext tasks are combined, it is too complicated for the optimization of a large number of parameters.

In terms of classification accuracy, although the proposed MEE training method is not comparable to the combined results of most of the best self-supervised auxiliary tasks, it can basically approach or surpass in a few cases. However, in the inductive setting, the best results of the combination of the self-supervised and self-distillation on UC Merced, WHU-RS19, and NWPU0RESISC45 are 0.55%, 0.53%, and 0.67% higher than those of MEE, respectively, while 0.55%, 0.31%, and 0.40% are improved in the transductive setting. However, in terms of time complexity, the time consumed by the exhaustive method of four self-supervised auxiliary tasks and one self-distillation shown in Table VIII is 0.2894, 0.3357, and 5.5267 GPU days in UC Merced, WHU-RS19,

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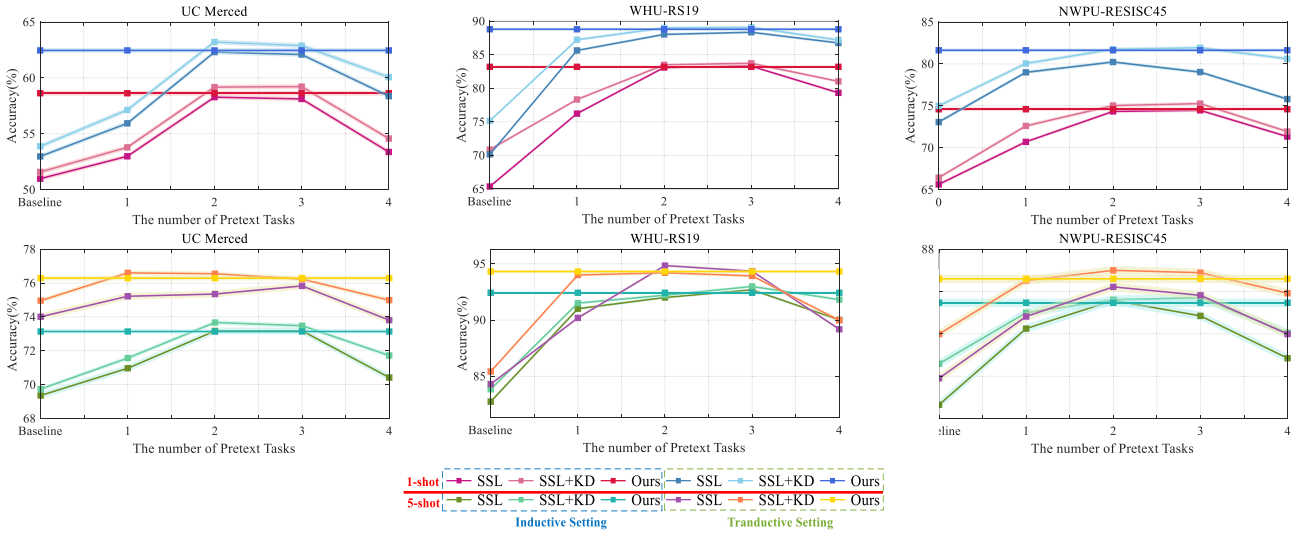


Fig. 9. **Accuracy comparison (in %) of the exhaustive method and the proposed MEE.** This figure only shows the accuracy of the best self-supervised combination for each fixed number of pretext tasks.

TABLE IX

OVERALL ACCURACY (IN %) OF COMBINING THE MES^{2}L FRAMEWORK WITH METALEARNING OR METRIC LEARNING METHODS ON THREE BENCHMARK SCENE CLASSIFICATION DATASETS

Method	UC Merced		WHU-19		NWPU-RESISC45	
	5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
MatchingNet (MN)	46.46 ± 0.75	67.39 ± 0.50	60.14 ± 0.67	68.66 ± 0.38	45.80 ± 0.86	64.35 ± 0.65
MES^{2}L + MN	$\mathbf{63.93 \pm 0.18}$	$\mathbf{80.91 \pm 0.11}$	$\mathbf{85.20 \pm 0.12}$	$\mathbf{91.52 \pm 0.05}$	$\mathbf{79.42 \pm 0.20}$	$\mathbf{89.86 \pm 0.13}$
Pretraining + MN	48.22 ± 0.70	70.03 ± 0.47	61.90 ± 0.61	70.10 ± 0.35	47.29 ± 0.80	67.04 ± 0.59
MES^{2}L$^{\mathcal{P}}$ + MN	55.30 ± 0.45	75.37 ± 0.32	72.49 ± 0.41	82.74 ± 0.22	59.60 ± 0.72	78.18 ± 0.43
Prototypical Net (PN)	54.03 ± 0.78	70.81 ± 0.48	69.03 ± 0.65	84.84 ± 0.33	62.43 ± 0.84	78.68 ± 0.55
MES^{2}L + PN	$\mathbf{64.01 \pm 0.18}$	$\mathbf{80.89 \pm 0.11}$	$\mathbf{85.15 \pm 0.12}$	$\mathbf{91.48 \pm 0.12}$	$\mathbf{79.35 \pm 0.20}$	$\mathbf{89.83 \pm 0.13}$
Pretraining + PN	56.28 ± 0.72	71.76 ± 0.48	70.84 ± 0.63	85.25 ± 0.32	63.88 ± 0.83	80.61 ± 0.55
MES^{2}L$^{\mathcal{P}}$ + PN	60.36 ± 0.49	74.77 ± 0.35	78.51 ± 0.42	87.43 ± 0.22	70.95 ± 0.70	84.83 ± 0.40
Relation Net (RN)	49.77 ± 0.78	57.98 ± 0.57	70.01 ± 0.68	86.04 ± 0.32	66.58 ± 0.91	76.11 ± 0.58
MES^{2}L + RN	55.21 ± 0.27	66.64 ± 0.25	74.48 ± 0.27	88.71 ± 0.24	70.81 ± 0.31	80.33 ± 0.29
Pretraining + RN	52.49 ± 0.76	59.20 ± 0.60	71.36 ± 0.68	87.09 ± 0.31	67.23 ± 0.90	77.09 ± 0.57
MES^{2}L$^{\mathcal{P}}$ + RN	58.52 ± 0.57	70.01 ± 0.38	79.92 ± 0.41	90.66 ± 0.27	70.15 ± 0.72	79.53 ± 0.45

and NWPU-RESISC45 datasets, respectively, while our proposed MEE correspondingly takes only 0.0232, 0.0289, and 0.5151 GPU days in these three remote sensing scene datasets. In addition, there are a total of ten combinations of the four self-supervised auxiliary tasks, plus a self-distillation operator, a total of 20 networks need to be trained, while the MES^{2}L-Net only needs to be trained once. This reflects that the MES^{2}L-Net can approach the classification performance of the best combination of self-supervised auxiliary tasks and self-distillation at a lower time cost.

E. Extensibility

The proposed MES^{2}L is not only limited to the transfer learning scheme but also can be combined with other few-shot algorithms based on metalearning and metric learning. Table IX presents the improved results before and after the combination of the MES^{2}L framework and the classic MN [53], PN [52], and RN [43]. "MES^{2}L+" refers to using the MES^{2}L framework to train the feature extractor and then using different metric learning methods to discriminate between the support set and the query set. As shown in Table IX, under the same metric learning strategy, the features extracted by MES^{2}L perform better in both accuracy and discriminability on the three backbone networks. Specifically, the improvement on MN is the most obvious with the largest improvement even reaching 33.62% in a 5-way 1-shot case on the WHU-RS19 dataset. This significant improvement is mainly due to the more discriminative features extracted by the MES^{2}L framework, so the samples in the query set can be well classified no matter which metric is employed.

In [10], [11], [12], [13], [14], and [15], the feature extractor is first trained with the based dataset, and then, the pretrained feature extractor is used as the initial feature extractor in metalearning. In the same way, the feature extractor trained with the MES^{2}L framework can be employed as the pretraining for metalearning methods, which is represented by "MES^{2}L$^{\mathcal{P}}$+." "Pretraining+" indicates that only the "Baseline" is used as pretraining for metalearning. The experimental results show that the MES^{2}L combined with pretraining for metatraining and metatesting has all been improved. Compared with pretraining the backbone with cross-entropy loss on the base

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dataset, the design of global–local contrastive learning and MEE strategy is more beneficial to improve the performance of the network trained with metalearning.

It is worth noting that the improvement of RN compared to PN or MN is not relatively ideal. The RN consists of an embedding model and a relation model. The feature maps of the images in the support set and the query set are extracted with the embedding model, and then, the dimension representing the number of channels in the feature map is spliced to obtain a new feature map. The new feature map is sent to the relation model for the operation to obtain the relation score, which represents the similarity between the two images, while we directly use the previously trained neural network for the relation model, and the metric space is largely dependent on the training data, so it does not perform well for the new input data.

V. Conclusion

In this article, we proposed an MES^{2}L framework for FSRSSC. Based on the transfer learning few-show manner, we design a novel global–local contrastive learning auxiliary task and an MEE training method in MES^{2}L-Net. In terms of the global–local contrastive learning, we address the low between-class separability by implementing the SAM in local contrastive learning. To obtain a tradeoff between time cost and classification accuracy, the MEE training method aims to replace the exhaustive combination of multiple pretext tasks and self-distillation by randomly selecting the self-supervised auxiliary tasks in multiple backbones. In addition, two classification schemes based on inductive and transductive settings are designed to solve real-life data acquisition and data labeling difficulties. Compared with a variety of excellent few-shot algorithms, the experimental result of MES^{2}L-Net is SOTA. Combining the MES^{2}L framework with classical metalearning or metric learning-based few-shot algorithms also demonstrates its universality and generalization performance.

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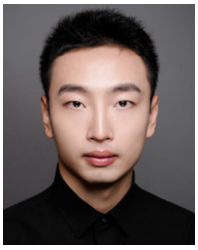
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